2024 Spring 生化学 2 English, Home work#3(250331)　　dead line: 23:59, April 7, 2025

学号:

姓名:

Q1. 判断题（请在括号里画√或 X, 1 分/题, 共 40 分）

(1) Passive transporter moves substrates against the electrochemical gradient. (　　)
(2) Transporter reduces the activation energy for transmembrane diffusion of the solute. (　　)
(3) With the symport system, two substrates (S1 and 2) are transported opposite direction. (　　)
(4) Glucose Transporter like GluT1 decrease the rate of glucose transport compare to that of uncatalyzed transmembrane diffusion. (　　)
(5) All Glucose Transporters (GluTs) have similar structure, therefore have similar kinetic value (e.g. K_t). (　　)
(6) In GluT, side-by-side assembly of several helices produces a transmembrane channel lined with hydrophilic residues. (　　)
(7) GluT uses ATP to transport glucose. (　　)
(8) In the Secondary active transport system, both primary and secondary transport use ATP. (　　)
(9) Energy cost of charged solute transport depends both the chemical gradient and electrical potential. (　　)
(10) P-type ATPase is irreversibly phosphorylated by ATP. (　　)
(11) V-Type ATPase use ATP to transport H^+. (　　)
(12) F-Type ATPase can synthesize ATP using the proton gradient. (　　)
(13) ABC transporter use GTP to drive the active transport. (　　)
(14) Some kind of multi drug transporters is ABC transporter. (　　)
(15) Some cotransport systems use chemical potential of ion (e.g. Na^+). (　　)
(16) Ionophore binds to ion to provide the hydrophilic exterior. (　　)
(17) Transport speed of ion channel is slower than ion pump (　　)
(18) Aquaporin can transport only water. (　　)
(19) K^+ channel can also transport smaller size ion like Na^+. (　　)
(20) Ion channel can transport against electrochemical gradient. (　　)
(21) Patch Clamp method measures the current change as the channel transits

between the open and closed state. ( )

(22) GPCRs directly activate the effector proteins like Adenylyl cyclase. ( )

(23) Rhodopsin accelerates the exchange from GTP to GDP of G-protein. ( )

(24) Hydrolysis speed of G-protein is slow, but some other proteins (e.g. GAP) accelerate the hydrolysis speed. ( )

(25) Adenylyl cyclase catalyzes the formation of cAMP from ADP. ( )

(26) cAMP activate protein kinase A. ( )

(27) Phospholipase A cleaves PIP2 to IP3 and diacylglyceryol. ( )

(28) IP3 binds to a specific receptor-gated Ca2+ pump. ( )

(29) Protein Kinase C (PKC) is activated by diacylglycerol ad Ca2+. ( )

(30) Ca2+/Calmodulin binding changes the structure of CaMK (Calmodulin - dependent protein kinase). ( )

(31) Some GPCR system use cGMP instead of cAMP. ( )

(32) Receptor Tyrosine Kinases can autophosphorylation, where several Asp are phosphorylated. ( )

(33) Receptor Tyrosine Kinases uses secondary messenger (e.g. cAMP) to activate the downstream signaling pathway. ( )

(34) Some proteins (e.g. MEK and ErK) in a protein kinase cascade can bind to scaffold protein for efficient regulation of the cascade. ( )

(35) Voltage-gated K+ channel transports the K+ ion from the outside to the inside of the axon. ( )

(36) In the cross talk of several pathways, Insulin receptor (INSR) can only regulate GPCR negatively. ( )

(37) Phosporylation of Thr is the key for CDK activation. ( )

(38) In regulation, ubiquitin ligase adds ubiquitin molecules to cyclin as the degradation tag. ( )

(39) Double-strand break in DNA sometimes works as the checkpoint from G1 to S phase. ( )

(40) Mutations in oncogenes and growth factors affect the normal cells to the cancer cells. ( )

Q2. Explain the five features of signal-transducing system (5 分)

Q3. Please explain the character of desensitizing compare to termination (2 分). Also, please explain the desensitizing mechanism of b-adrenergic receptor(3 分)